

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**3<sup>rd</sup> Semester of B. Pharm. Examination**  
**University Theory Examination April 2016**  
**PH205 Pharmaceutical Engineering-III**

**Date: 18-04-16, Monday**

**Time: 1:30 p.m to 4:30p.m**

**Maximum Marks: 80**

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**Instructions:**

- 1. Figures to right indicate marks.**
- 2. Section wise separate answer book to be used.**
- 3. Draw neat sketches wherever necessary.**

**SECTION- I**

- Q 1** Attempt any **TWO** of the following
- A a) Explain various mechanisms of size reduction along with examples for each. **05**  
b) Explain various application of size reduction. **05**
- B Describe with a clean and labelled diagram the construction, principle, working, advantages, disadvantages and application of cyclone separator. List the specifications and standards for sieves. **10**
- C a) Define and explain the following: **10**  
1). Mixing 2). Mixing index  
b) Explain in detail the Planetary mixer.
- Q 2** Attempt any **TWO** of the following
- A a) Draw diagram of Ball mill and Fluid energy mill respectively **05**  
b) Explain any one theory related to the size reduction of a powder. **05**
- B Explain the mechanisms of solid -liquid mixing. Explain principle, construction, applications, advantages and disadvantages of double cone mixer **10**
- C Write application of size separation. Explain any one separation tank in detail **10**

**SECTION- II**

- Q 3** Attempt any **TWO** of the following
- A Discuss in detail challenges with particle engineering approaches. **10**
- B Write a note on Mier's super saturation theory. Discuss its limitations and applications. **10**
- C Classify crystallizers. Write a detailed note on construction, principle, working, advantages, disadvantages and application of any one. **10**
- Q 4** Attempt any **TWO** of the following
- A Discuss in detail the elements of automated process control systems. **10**
- B Write a detail note on automated process control system for temperature control. **10**
- C Discuss fire and safety hazards with respect to industrial hazard and safety precautions. **10**
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